

Methodology: Intelligent Candidate Discovery

Objective

Rank the top 100 professionals for Redrob AI's Senior AI Engineer role. The job description asks for a practical builder who has shipped production retrieval, ranking, matching, and evaluation systems, not simply someone with AI keywords in a skills list.

Architecture

The system is a deterministic, CPU-only ranking pipeline:

1. Stream 'candidates.jsonl' one record at a time.
2. Build a compact evidence text from the profile, career history, industries, titles, and skills.
3. Score each candidate across explainable buckets.
4. Keep only the top 100 candidates in a heap for memory efficiency.
5. Emit the official CSV plus an XLSX copy with the same columns.

No hosted LLM APIs, GPU inference, or network calls are used during ranking.

Scoring Buckets

1. Role And Seniority Fit

Candidates receive strong positive weight for AI, ML, NLP, search, ranking, applied scientist, backend, data, and senior engineering titles. The experience curve favors roughly 5-9 years, with the strongest region around 6-8 years, matching the JD's stated intent.

2. Production Retrieval And Ranking Evidence

The highest-weight signals come from career descriptions, not from isolated skills. The system rewards evidence such as production ranking pipelines, hybrid retrieval, semantic search, BM25 plus dense recall, vector search, NDCG/MRR/MAP, A/B tests, recruiter-facing search, recommendation systems, candidate-JD matching, and operational scale.

3. Skills Inventory

The ranker scores skills based on relevance, proficiency, duration, and endorsements. High-value skills include Python, NLP, information retrieval, embeddings, semantic search, vector databases, FAISS, Pinecone, Weaviate, Qdrant, Milvus, OpenSearch, Elasticsearch, BM25, learning-to-rank, recommendation systems, and LLM fine-tuning.

4. Career Context

The JD explicitly values product-building judgment. The system therefore boosts candidates with product-company, AI/ML, marketplace, fintech, SaaS, e-commerce, or startup exposure. It penalizes consulting-only backgrounds when there is no product engineering evidence.

5. Behavioral Availability

Behavioral signals adjust the profile fit score. Recent activity, open-to-work status, recruiter response rate, short notice period, interview completion rate, profile completeness, recruiter saves, and GitHub activity all improve confidence that the candidate is hireable now.

6. Risk And Trap Handling

The dataset contains keyword stuffing and honeypot-style profiles. The system down-weights non-engineering titles that merely mention AI keywords, course-only or side-project-only AI exposure, stale profiles, long notice periods, CV/speech/robotics-only profiles without IR/NLP/ranking evidence, and implausible skill-duration patterns.

Explainability

Every output row includes a 1-2 sentence reason using facts from the candidate profile: current title, years of experience, location, matched skills, production evidence, response rate, notice period, and any relevant concern.

Compute Fit

The implementation streams JSONL and keeps a 100-item heap, so memory usage is small relative to the 16 GB limit. The ranker uses only Python's standard library and avoids network calls, GPU dependencies, or per-candidate LLM calls.

Expected Strength

The approach is designed to avoid the main challenge trap: ranking candidates by keyword overlap alone. Career-history evidence and behavioral availability have meaningful weight, while isolated skills or recent AI curiosity do not dominate the score.